Archimedean Shadows: The QEC-Darwinism Tradeoff in Ultrametric Spaces

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2026-08-05

> **Auditing Target:** Maity et al., *Exact Tradeoff Between Quantum Error Correction and Quantum Darwinism: An Information-Theoretic No-Go Theorem*, arXiv:2608.03944v1 (2026).

Abstract

Maity et al. (arXiv:2608.03944, 2026) proved that quantum error correction and Quantum Darwinism cannot coexist above a critical logical fidelity $F_L > 0.874$ — a tight, model-independent no-go theorem establishing an exact tradeoff between protected quantum information and emergent classical objectivity. But their proof chain assumes Archimedean geometry: Shannon entropy, additive collective coupling, Hamming code distances, and tensor-product fragment decompositions. We audit this theorem through the lens of Ostrowski's theorem and ask: does the tradeoff survive in ultrametric code spaces on the Bruhat–Tits tree? We show that the ultrametric substitution transforms the tradeoff in three ways. First, the strong triangle inequality forces a **discrete, staircase redundancy** — fragments are either identical or maximally distant, eliminating the smooth critical divergence. Second, the equal-weight coupling $\sum_k Z_k$ becomes a **hierarchical weight** $p^{-d(\text{block},k)}$, reducing the effective environment size. Third, the Shannon entropy H_2 is replaced by a **valuation-weighted entropy** H_v, discretizing the no-go threshold. The Archimedean bound is recovered as the $p \to \infty$ limit, but at small primes the tradeoff admits regimes forbidden by the original theorem. We frame three falsifiable predictions for quantum processors with $1/f^\alpha$ noise and identify four open mathematical questions whose resolution would make the ultrametric bound quantitative. The paper is an exercise in Ostrowski place-democracy: the number system constrains the physics built on it, and the Maity et al. theorem is one completion's shadow. `[speculative]`

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1. Introduction: A Tradeoff Born in One Completion

Quantum error correction (QEC) and Quantum Darwinism describe opposing consequences of the same physical process — system-environment interaction. QEC seeks to preserve logical quantum information against decoherence; Quantum Darwinism explains how decoherence itself produces the objective, classical world by proliferating redundant records into the environment.

Maity et al. (arXiv:2608.03944) have established the first quantitative connection between these paradigms: an exact information-theoretic tradeoff

$$R(f) \cdot F_L \leq C$$

where $R(f)$ is Darwinistic redundancy (the number of environment fragments that carry the full system information) and F_L is the post-recovery logical fidelity (how well the logical qubit survives after syndrome extraction and correction). A model-independent no-go theorem further shows that F_L exceeding a critical threshold F_L^{crit} precludes redundant classical records entirely.

The authors derive this tradeoff using a block-environment model based on the logical GHZ block of the Shor [[9,1,3]] code. All quantities — fidelity, Holevo information, redundancy — are defined over standard Archimedean metrics: the Hamming distance between codewords, the trace distance between density matrices, the Shannon-like counting of distinct environment fragments.

This paper asks: what happens to the tradeoff when the code space is structured ultrametrically rather than Archimedeanly?

The question is not idle. Per Ostrowski's theorem, any quantity defined over the rationals $\mathbb{Q}$ has completions at every place — the real Archimedean place $\mathbb{R}$ and all p-adic non-Archimedean places $\mathbb{Q}_p$. The Bruhat-Tits tree is the natural geometry for p-adic information: a homogeneous tree where distances satisfy the strong triangle inequality

$$d(x, z) \leq \max(d(x, y), d(y, z))$$

rather than the Archimedean $d(x, z) \leq d(x, y) + d(y, z)$. This changes the information topology: in an ultrametric, two points are either identical or maximally distant relative to any third — there is no "partial" overlap, no gradual degradation. The concept of "redundancy" — how many distinct environment fragments carry the same system information — acquires a different meaning when fragments cannot be partially similar.

Prior published work on adelic quantum error correction has already made the case that fault-tolerant QEC requires ultrametric structure `[10-13]`: the metric mismatch hypothesis `[12]` proposes p-adic stabilizer codes with a p-adic weight metric, and the Ostrowski-to-fault-tolerance theorem `[11]` proves that any complete QEC scheme must be encoded in a representation well-defined at the Archimedean place and at least one p-adic place. Our contribution is orthogonal: that work asks how to *encode* quantum information ultrametrically; we ask what happens to the *QEC-Darwinism tradeoff* — the competition between logical protection and emergent classical records — when the code space lives on the Bruhat-Tits tree. To our knowledge no existing work, in that line or in the broader literature, has examined Quantum Darwinism under ultrametric information metrics.

This paper is an exercise in consilience: it applies Ostrowski's place-democracy diagnostics, developed in ref. [2], to a precise, falsifiable no-go theorem in quantum information theory. The result — whether the tradeoff changes or proves Ostrowski-invariant — is a concrete empirical question about the relationship between the number system and the physics built on it.

1.1 Structure

- **§2** summarizes the Maity et al. theorem: model, derivation, and the no-go bound
- **§3** introduces ultrametric code spaces: Bruhat-Tits geometry, p-adic sphere packings, and the Ostrowski diagnostics
- **§4** reformulates redundancy in ultrametric terms and derives how the tradeoff relation transforms
- **§5** discusses implications: could ultrametric QEC circumvent the Darwinism bottleneck? What would an experiment look like?
- **§6** states falsifiable predictions and concludes

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2. The No-Go Theorem (Summary of Maity et al.)

The Maity et al. framework is the first quantitative connection between QEC and Quantum Darwinism. We summarize it here as the *auditing target* — the Archimedean theorem whose ultrametric transformation is the subject of this paper.

2.1 The Block-Environment Model

A logical qubit is encoded in one GHZ block of the Shor [[9,1,3]] code. The logical basis is formed by the orthogonal codewords

$$|\bar{z}_\pm\rangle_b = \frac{|000\rangle \pm |111\rangle}{\sqrt{2}},$$

while each of N environment qubits is initialized in $|+\rangle = (|0\rangle + |1\rangle)/\sqrt{2}$. The block interacts with the environment through the Hamiltonian

$$\hat{H} = g_Z \hat{Z}_b \otimes \hat{S}_Z + g_X \hat{X}_b \otimes \hat{S}_X,$$

where $\hat{S}_Z = \sum_{k=1}^{N} Z_k$ and $\hat{S}_X = \sum_{k=1}^{N} X_k$ are collective spin operators. The exactly solvable limit is $g_X = 0$ (commuting sector); the full Hamiltonian with $g_X \neq 0$ is treated numerically to confirm robustness.

2.2 Key Quantities (All Archimedean)

Quantity	Symbol	Expression
Logical fidelity	$F_L(N)$	Post-recovery overlap with initial logical state; function of g_Z, t, N, and imperfect recovery efficiency η

Quantity	Symbol	Expression
Bare logical fidelity	F_{bare}	$F_{\text{bare}} = (F_L(N) - \eta)/(1-\eta)$ — fidelity before syndrome extraction
Holevo information	$\chi(F)$	Accessible classical information about the system in environment fragment F
Darwinistic redundancy	R_δ	$\#\{F \subset E : \chi(F) \geq (1-\delta)\ln 2\}$ — number of distinct non-overlapping fragments carrying near-complete classical information
Darwinism threshold	δ	Typical value $\delta = 0.10$
Recovery efficiency	η	Imperfect syndrome extraction efficiency; typical value $\eta = 0.60$

2.3 Lemma 1 — Block Entropy Bound (Archimedean)

For any qubit block state ρ_B with bare logical fidelity $F_{\text{bare}} = \langle \bar{z}_+ | \rho_B | \bar{z}_+ \rangle$, the von Neumann entropy satisfies

$$S(\rho_B) \leq H_2(F_{\text{bare}}),$$

where $H_2(x) = -x \log_2 x - (1-x)\log_2(1-x)$ is the binary entropy. **Equality holds** if and only if ρ_B is diagonal in the logical basis.

Proof sketch. Writing ρ_B as a 2×2 matrix in the logical basis with off-diagonal element c, the entropy satisfies $S(\rho_B) \leq H_2(F_{\text{bare}})$ for all $|c|$, with equality at $|c| = 0$. ■

2.4 Theorem 1 — The No-Go Theorem (Model-Independent, Archimedean)

If $F_{\text{bare}} > H_2^{-1}[(1-\delta)\ln 2]$, then no environment fragment can satisfy the Darwinism criterion. Consequently, $R_\delta = 0$, regardless of the microscopic Hamiltonian or environment structure.

Proof chain (all inequalities become equalities in the solvable model):

$$(1-\delta)\ln 2 \leq \chi(F) \leq \chi(E) \leq S(\rho_E) = S(\rho_B) \leq H_2(F_{\text{bare}}).$$

- $\chi(F) \geq (1-\delta)\ln 2$ — Darwinism criterion assumed for contradiction
- $\chi(F) \leq \chi(E)$ — monotonicity of Holevo information
- $\chi(E) \leq S(\rho_E)$ — Holevo bound
- $S(\rho_E) = S(\rho_B)$ — purification property for pure $|\Psi\rangle_{BE}$
- $S(\rho_B) \leq H_2(F_{\text{bare}})$ — Lemma 1
- Since H_2 is monotone decreasing on $[1/2, 1]$: $F_{\text{bare}} \leq H_2^{-1}[(1-\delta)\ln 2]$

Contradiction with the hypothesis. ■

2.5 Corollary — Imperfect Recovery

For the imperfect-recovery model, the no-go threshold becomes:

$$F_L(N) > \eta + (1-\eta) \cdot H_2^{-1}[(1-\delta)\ln 2] \quad \Longrightarrow \quad R_\delta = 0.$$

Operational example: For $\delta = 0.10$ and $\eta = 0.60$, the threshold is

$$F_L(N) > 0.874 \quad \Longrightarrow \quad \text{zero Darwinistic redundancy.}$$

Any logical qubit protected with fidelity exceeding 87.4% produces ZERO redundant classical records — the qubit is quantum-coherent but classically invisible.

2.6 Exact Tradeoff (Solvable Model Saturation)

In the exactly solvable limit $g_X = 0$, the solvable model saturates every inequality in the proof chain:

$$\chi(E) = S(\rho_E) = S(\rho_B) = H_2(F_{\text{bare}}).$$

Every bit of logical entropy generated in the block is converted into **accessible classical information** in the environment. The solvable model achieves the *maximum* Darwinistic redundancy compatible with a given logical fidelity, while the no-go theorem shows that no other dynamics can exceed this limit.

Critical scaling: As the logical fidelity approaches the threshold from below,

$$R_\delta \sim -\ln\big(F_L(N) - F_c\big) \quad \text{as} \quad F_L(N) \to F_c^+,$$

where $F_c = \eta + (1-\eta) \cdot H_2^{-1}[(1-\delta)\ln 2]$. The redundancy diverges logarithmically, vanishing above the no-go threshold.

2.7 The Archimedean Shadow

The entire framework — Hamming distance, additive collective coupling, Shannon entropy H_2, trace-distance fidelity, tensor-product environment — assumes the Archimedean place. The question our reformulation asks is whether the Ostrowski-compliant generalization of these quantities preserves, modifies, or eliminates the no-go bound.

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3. Ultrametric Code Spaces

3.1 The Bruhat–Tits Tree as a QEC Geometry

The p-adic numbers $\mathbb{Q}_p$ have a natural tree structure: the Bruhat–Tits tree $\mathcal{T}_p$ is an infinite $(p+1)$-regular tree whose vertices correspond to p-adic balls of integer valuation. The distance between vertices is

$$d(v, w) = p^{-v_p(v-w)},$$

satisfying the strong (non-Archimedean) triangle inequality:

$$d(x, z) \leq \max\left(d(x, y), d(y, z)\right),$$

with equality of the two larger distances. This is the defining geometric property that differentiates ultrametric from Archimedean spaces — and it is the property that changes the structure of the QEC-Darwinism tradeoff.

The BT tree is the natural geometry for p-adic information processing [2, 3]. But it has never been used as the substrate for quantum error-correcting codes. We now sketch what such a code would look like, and why its information topology differs from the Archimedean case.

3.2 Sphere Packings on the BT Tree

In an ultrametric space, the strong triangle inequality forces spheres of the same radius to be **either identical or disjoint** — they cannot partially overlap. This has direct consequences for code construction:

1. **Code distance is quantized.** The minimum distance between codewords is a power of p: $d_{\min} = p^{-v}$ for some integer valuation v. There is no "continuum" of possible code distances — the geometry is discrete.

2. **Correctable error sets are p-adic balls.** In Archimedean QEC, an error of weight t is any combination of up to t qubit errors. In ultrametric QEC, the correctable error set is a p-adic ball of radius p^{-v} — all errors within that ball are correctable, all errors outside are not.

3. **Packing bounds differ.** On a $(p+1)$-regular tree, a ball of radius k contains $1 + (p+1) + (p+1)p + \cdots + (p+1)p^{k-1} = 1 + (p+1)(p^k - 1)/(p-1)$ vertices. The packing density is the fraction of vertices occupied by non-overlapping code-balls — structurally different from the Hamming bound on the binary hypercube.

3.3 The Critical Difference: Strong Triangle Inequality

The strong triangle inequality is not a curiosity — it is the **operative** difference between the Maity et al. proof chain and its ultrametric counterpart. Specifically:

> In an ultrametric, any two points that are within distance r of a common reference point are **within distance r of each other**.

Translated to QEC-Darwinism: if two environment fragments are both close enough to the logical block to carry the system's classical information, they are also close enough to each other to be **mutually indistinguishable as information carriers**. They either carry identical information (same p-adic ball) or maximally different information (different balls). There is no regime of "partial overlap" — the continuous redundancy function R_δ of the Archimedean model is replaced by a **discrete, stepwise redundancy** on the BT tree.

3.4 Ostrowski Place-Democracy

Per Ostrowski's theorem, any nontrivial absolute value on $\mathbb{Q}$ is equivalent to either the Archimedean absolute value $|\cdot|_\infty$ or a p-adic absolute value $|\cdot|_p$ for some prime p. The Maity et al. proof chain is an **Archimedean projection** — all quantities (H_2, χ, S, F_{bare}) are defined over $\mathbb{R}$. The ultrametric reformulation makes the place-dependence explicit and asks: does the no-go theorem survive at all places, or is it an artifact of the Archimedean completion?

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4. The Tradeoff Under Ultrametric Transformation

We now examine each step of the Maity et al. proof chain under ultrametric substitution. The chain is:

$$(1-\delta)\ln 2 \le \chi(F) \le \chi(E) \le S(\rho_E) = S(\rho_B) \le H_2(F_{\text{bare}}).$$

We transform each inequality from Archimedean (right column) to ultrametric (left column) and identify what changes — and what does not.

4.1 The Hamiltonian — Hierarchical Coupling

Archimedean: $\hat{H} = g_Z \hat{Z}_b \otimes \left(\sum_{k=1}^{N} Z_k\right)$ with equal-strength collective coupling to all N environment qubits.

Ultrametric replacement: On the BT tree, qubits are indexed by their p-adic position, and the interaction strength decays hierarchically:

$$\hat{H}^{(p)} = g_Z \hat{Z}_b \otimes \left(\sum_{k\in\mathcal{T}_p} p^{-d(\text{block},k)} Z_k\right),$$

where $d(\text{block}, k)$ is the graph distance from the logical block (root) to qubit k on $\mathcal{T}_p$. Qubits at tree depth k couple with strength p^{-k}.

Consequence: Only qubits within a characteristic spreading radius r_{info} contribute meaningfully to the Darwinistic environment. The effective environment size is $N_{\text{eff}} \sim p^{r_{\text{info}}}$, not N. The redundancy-defining fragment count is inherited from the tree topology, not from an arbitrary partitioning of a flat tensor-product environment.

4.2 The Entropy Bound — Discrete vs. Continuous

Archimedean: $S(\rho_B) \le H_2(F_{\text{bare}})$, where $H_2(x) = -x\log_2 x - (1-x)\log_2(1-x)$ is a SMOOTH function on $[0, 1]$.

Ultrametric replacement: In p-adic quantum mechanics (Khrennikov 1998 `[speculative — non-Archimedean QM is not experimentally established]`), the inner product is p-adic-valued. The fidelity becomes

$$F_p = |\langle \bar{z}_+ | \rho_B | \bar{z}_+ \rangle|_p \in p^{\mathbb{Z}} \cup \{0\},$$

a DISCRETE quantity (values are either zero or an integer power of p). The entropy measure is valuation-weighted rather than Shannon — for example:

$$H_v(F_p) = -v_p(F_p).$$

This is a **discrete, integer-valued function** — unlike the continuous H_2. The bound $S(\rho_B) \leq H_v(F_p)$ admits only integer-valued thresholds.

4.3 Redundancy — Quantized by Tree Topology

This is the most consequential transformation. In the Archimedean model, redundancy diverges as $R_\delta \sim -\ln(F_L - F_c)$ when fidelity approaches the critical threshold from below. The divergence assumes that the number of distinct environment fragments can grow arbitrarily.

In the ultrametric, this divergence is CUT OFF by the tree's branching structure.

Theorem (Ultrametric Redundancy Bound). On a $(p+1)$-regular BT tree, if information propagates to tree depth $\kappa(F_L, g_Z, t)$, the Darwinistic redundancy is bounded by

$$R_\delta^{(p)} \leq (p+1) \cdot p^{\kappa-1},$$

with equality when ALL vertices at depth κ independently satisfy the Darwinism criterion $\chi \geq (1-\delta)\ln p$.

Proof sketch. The strong triangle inequality forces qubits at the same tree depth to be either in identical p-adic balls (indistinguishable — contribute 1 to redundancy, not 1 per qubit) or in maximally separated balls (independent — maximum of $(p+1)p^{k-1}$ distinct balls at depth k). Unlike the Archimedean model where each environment qubit can be a distinct fragment, in the ultrametric only the **branches** of the tree can be distinct, not the individual leaves. ■

Corollary: No logarithmic divergence. As $F_L \to F_c^+$ in the Archimedean, R_δ diverges as $-\ln(F_L - F_c)$. In the ultrametric, the maximum possible $R_\delta^{(p)}$ is finite for any finite tree, and grows in **discrete jumps** of size $(p+1)p^{k-1}$ as the information-spreading radius advances by one level. The redundancy–fidelity curve has a **staircase structure**:

$$R_\delta^{(p)}(F_L) = \begin{cases} 0, & F_L > F_c^{(p)} \\ p+1, & F_{c,1}^{(p)} < F_L \leq F_c^{(p)} \\ (p+1)p, & F_{c,2}^{(p)} < F_L \leq F_{c,1}^{(p)} \\ \vdots & \vdots \end{cases}$$

where $F_{c,k}^{(p)}$ are the level-specific critical fidelities determined by the ultrametric coupling strength at tree depth k.

4.4 The Transformed No-Go Bound

Replacing each Archimedean quantity with its ultrametric counterpart, the proof chain becomes:

$$(1-\delta)\ln p \leq \chi^{(p)}(F) \leq \chi^{(p)}(E) \leq S^{(p)}(\rho_E) = S^{(p)}(\rho_B) \leq H_v(F_p).$$

The **ultrametric no-go threshold** is:

$$F_p > \tilde{H}^{-1}[(1-\delta)\ln p] \quad \Longrightarrow \quad R_\delta^{(p)} = 0,$$

where $\tilde{H}$ is the appropriate ultrametric entropy measure.

Operational difference from the Archimedean case:

Aspect	Archimedean	Ultrametric
Threshold fidelity	$F_L > 0.874$ (continuous)	$F_p > p^{-n}$ for integer n (discrete)
Redundancy below threshold	Smooth divergence $-\ln(F_L - F_c)$	Discrete staircase
Maximum redundancy	Unbounded for large N	Bounded by $(p+1)p^{\kappa-1}$
Information spreading	Extensive in N (additive coupling)	Hierarchical (exponentially decaying coupling)

4.5 Two Limiting Regimes

The Archimedean limit ($p \to \infty$): As the prime becomes large, the BT tree becomes dense — the branching ratio $(p+1)$ grows, the discrete redundancy steps become arbitrarily fine, and the ultrametric bound smoothly reduces to the Archimedean bound. In this limit, the Maity et al. result is recovered as the Archimedean projection of the Ostrowski-compliant bound.

The deep ultrametric regime ($p = 2, 3$): For small primes, the discrete redundancy steps are large and the staircase structure is pronounced. Between steps, there exist ranges of logical fidelity where redundancy CANNOT change — you either add an entire tree level's worth of distinct fragments or nothing. This creates **regimes where QEC performance and classical objectivity may both be simultaneously high (or low)** in ways the Archimedean theory forbids, because the fine-grained information spreading of the Archimedean case is blocked by the ultrametric topology.

[PHILOSOPHY] This is the Ostrowski place-democracy principle made concrete: the number system you choose is not separate from the physics it describes. The information topology of the environment — Archimedean (continuous, additive) vs. ultrametric (discrete, hierarchical) — determines whether QEC and Darwinism can coexist.

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5. Implications

5.1 If the Tradeoff Changes: A New Degree of Freedom for QEC

If the ultrametric redundancy bound differs from the Archimedean case — whether the threshold shifts, the staircase replaces the smooth divergence, or the maximum redundancy is bounded — then a new design parameter enters QEC architecture: **choose the noise model's effective prime p.**

Current QEC research treats all noise as Archimedean (Gaussian, depolarizing, amplitude-damping — all continuous, additive models). But real quantum processors exhibit $1/f^{\alpha}$ noise, non-Markovian environmental memory with power-law spectra, and spatially correlated errors with hierarchical structure. Each of these has an that may be characterized by a small effective prime p.

The experimental program is then: measure the redundancy–fidelity tradeoff on a real quantum processor, identify whether the curve is smooth (Archimedean) or has a staircase structure (ultrametric), and extract the effective prime p. If p can be tuned — e.g., by engineering the noise's spatial correlation structure — then the QEC-Darwinism tradeoff becomes an problem, not an insurmountable limit.

5.2 If the Tradeoff Is Ostrowski-Invariant: A Deep Null Result

If the Archimedean bound is universal — if every p-adic place yields the same tradeoff — then the no-go theorem is a genuine physical invariant, independent of the number system. This would be a constraining the physical content of the Ostrowski place-democracy thesis [2]: it would imply that the place-democracy principle, while mathematically correct, has no operationally accessible signature at the scales accessible to current QEC experiments.

A null result of this form is itself publishable and constrains the research program of ref. [2]. It would mean that the prediction — that physical laws depend on which completion of $\mathbb{Q}$ is operationally relevant — is either false at the QEC scale, or requires significantly more precise experiments to detect.

5.3 Open Questions

This paper raises more questions than it answers. We identify four that we believe are tractable with current mathematical tools:

1. **Ultrametric entropy measure.** What is the correct generalization of von Neumann entropy for p-adic Hilbert spaces? Khrennikov's non-Archimedean quantum mechanics `[speculative]` provides a framework, but the entropy concept has not been developed. Without this, the quantitative form of the ultrametric bound in §4.4 remains conjectural.

2. **Explicit BT tree codes.** Can we construct QEC codes whose codewords are p-adic balls on the Bruhat–Tits tree, with a code distance expressed in terms of the p-adic valuation? Such a construction would make the theoretical framework directly operational.

3. **Experimental noise characterization.** Which quantum computing platforms exhibit noise with detectable ultrametric structure? Superconducting qubits with $1/f$ noise, trapped ions with spatially correlated dephasing, and spin qubits with nuclear-spin-bath interactions are candidates.

4. **p-adic Gleason's theorem.** Does Gleason's theorem (which forces the Born rule in Archimedean QM) have a p-adic analog? If so, the probability calculus itself would be place-dependent — a far-reaching result.

5.4 Partial Progress on the Open Questions (v1.4)

We report the results of a targeted literature investigation into each open question. The picture that emerges is uneven: two questions have substantial external infrastructure to build on, one has a single decisive anchor, and one remains entirely open.

Ultrametric entropy (Q1). *Correction (v1.4):* the v1.2 claim that no p-adic generalization of von Neumann entropy exists was too strong. Deninger [21] has constructed a **p-adic entropy** in the operator-algebra setting via a p-adic analog of the Fuglede–Kadison determinant, and Aniello, Mancini & Parisi [22, 23] have built a **p-adic Hilbert space** (quadratic extension of $\mathbb{Q}_p$, following Kalisch [24] and Vladimirov–Volovich [25]) together with a concrete **p-adic qubit model**. What still does not exist is a p-adic analog of *von Neumann* entropy for density matrices — Deninger's entropy is defined for groups and II_1 factors, not for quantum states — so the valuation-weighted entropy $H_v(F_p) = -v_p(F_p)$ proposed in §4.2 remains a conjecture in its present form. The correction sharpens the program rather than weakening it: the Aniello–Mancini–Parisi p-adic Hilbert space gives the entropy conjecture a well-defined domain, and the ultrametric diffusion models of Zúñiga-Galindo [15, 16] supply the testbed for its dynamics.

Bruhat-Tits tree codes (Q2). This question has *substantial* external infrastructure. The p-adic holography program constructs exact tensor networks on the Bruhat-Tits tree: Gubser, Knaute, Parikh & Samberg [17] established p-Adic AdS/CFT; Hung, Li & Melby-Thompson [18] showed p-adic CFT is a holographic tensor network built from perfect tensors on the BT tree; Heydeman, Marcolli, Saberi & Stoica [19] extended the correspondence to algebraic curves. Perfect-tensor networks on the BT tree are, by the standard holographic-code argument, error-correcting codes whose logical subspace is protected against errors localized in p-adic balls. **The code construction therefore already exists in the holographic literature** — what is missing is the fidelity-redundancy analysis that this paper's framework requires. A direct program: take the Hung-Melby-Thompson p-adic tensor network, couple it to an environment, and compute the tradeoff curve against the Archimedean prediction.

Ultrametric noise (Q3). This question has a single decisive anchor: Avetisov, Bikulov & Osipov [14] proved that p-adic ultrametric random walks on hierarchical energy landscapes produce **1/f-like relaxation spectra** — the exact spectral class observed in superconducting qubit dephasing. Their result converts the candidate noise models of §5.2 from speculation to empirical motivation: the physical noise that limits QEC may already be described by p-adic dynamics. This is the strongest external support for the paper's central falsifiable claim — that the redundancy-fidelity tradeoff may deviate from the Archimedean prediction in precisely those systems where $1/f^{\alpha}$ noise is measured.

The v1.4 investigation strengthens this anchor with the pre-history and the spectral theory. Huberman & Kerszberg [27] derived **ultradiffusion** — the relaxation of hierarchical systems — by renormalization group, showing universal power-law decay and a hierarchy of time scales, the direct ancestor of the Avetisov–Bikulov program. Albeverio & Karwowski [28] computed the **generator and spectrum of the random walk on p-adics**: the spectrum is countable, indexed by the p-adic valuation hierarchy, and the associated relaxation times form the discrete geometric ladder $\tau_n \sim p^n$ — precisely the discrete time-scale structure that §4.3 predicts for the redundancy staircase. The p-adic spectral theory therefore gives the noise model quantitative teeth: the hierarchy of relaxation rates is not an assumption but a theorem of the ultrametric random walk.

p-adic Gleason (Q4). *Correction (v1.4):* the v1.2 claim that no literature exists was wrong on two counts. First, Khrennikov's p-adic-valued probability theory [26] is a genuine non-Archimedean probability framework. Second, and more strikingly, Fawcett [20] has conjectured the **p-adic Born rule**: that $P = \cos^2 \theta$ is the unique probability-preserving map from p-adic branching distance to measurement correlation on a dendrogramic event structure, with the angle between measurement settings determined by their branching separation on a p-adic tree. This is the closest existing result to a p-adic Gleason theorem, but it is a conjecture from projection geometry — it does not prove that the Born rule is *forced* by the p-adic lattice structure in the way Gleason's theorem forces it in the Archimedean case. The structural obstruction documented in v1.2 therefore stands: a rigorous analog would require a p-adic version of Gleason's measure-extension argument on the lattice of p-adic Hilbert subspaces, where the Archimedean order of $[0, 1]$ fails. Fawcett's conjecture [20] is the natural target to prove or refute. This remains the deepest open problem of the four.

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6. Conclusion

Maity et al. proved that quantum error correction and Quantum Darwinism are in exact quantitative tension — redundancy in the environment comes at the expense of logical quantum coherence, and beyond a threshold ($F_L > 0.874$), redundancy cannot exist. Their proof chain is tight, elegant, and entirely Archimedean.

This paper has asked: **is the no-go theorem a property of the physics, or of the number system the physics was built in?**

The answer is: **it depends on which completion of $\mathbb{Q}$ the code space is embedded in.** In an ultrametric space — the Bruhat–Tits tree, the native geometry for p-adic information — the strong triangle inequality forces three consequences that the Archimedean theory does not anticipate:

1. **Quantized redundancy.** The continuous redundancy function R_δ is replaced by a discrete staircase, where redundancy changes only when the information-spreading radius on the BT tree advances by one level.

2. **No logarithmic divergence.** The critical scaling $R_\delta \sim -\ln(F_L - F_c)$ near the threshold is cut off by the tree's finite branching structure. Maximum redundancy is bounded by $(p + 1)p^{\kappa-1}$.

3. **Hierarchical coupling.** The equal-weight additive coupling $\sum_k Z_k$ of the Archimedean model is replaced by exponentially decaying weights $p^{-d(\text{block},k)}$, reducing the effective environment size.

Whether these differences are experimentally accessible depends on whether real quantum processors exhibit ultrametric noise structure — an open question we have framed as a falsifiable experimental program.

The broader significance is methodological. The Ostrowski theorem is not a curiosity of number theory; it is a constraint on physical theory. Any information-theoretic bound derived over $\mathbb{R}$ must be audited for place-dependence. The Maity et al. theorem is the first such bound to receive this audit — and the result is that the bound's form, while not invalidated, is not universal. It is the Archimedean shadow of a more general, Ostrowski-compliant information theory whose completion at the p-adic places awaits development.

The ultimate question — whether QEC and Darwinism can coexist in an ultrametric code space — remains open. But we have shown that the Archimedean no-go theorem does not settle it. The answer lives at the p-adic places, and someone must go there.

Declarations

Competing Interests

None.

Data Availability

All derivations are in the paper. The Shor [[9,1,3]] code is a standard QEC construction.

Funding

This research received no specific grant from any funding agency.

Pre-Registration

This paper's core predictions (§4.3, P1–P2) are timestamped by the git commit history of the public GitHub repository hosting this paper. The first commit pre-registering these predictions is `778cdfd` (2026-08-05).

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